

João de Teixeira da Encarnação

Résumé

Research Scientist Associate, Center for Space Research, University of Texas at Austin

Personal Information

Full Name: João Gregório de Teixeira da Encarnação
Birth: 25th of February 1977 at Funchal, Portugal
Nationality: Portuguese
Address: Klyuwerweg 1
2629 HS, Delft
the Netherlands
Telephone:
Email: j.g.deTeixeiradaEncarnacao@tudelft.nl
Web: [University of Texas](#), [TU Delft](#), [LinkedIn](#), [ResearchGate](#), [Google Scholar](#), [ORCID](#), [Mendeley](#), [SCOPUS](#), [Publons](#), [GitHub](#)

Education

- 2015 **Ph.D. in Space Geodesy**
Geoscience and Remote Sensing ([GRS](#)), Delft University of Technology ([TU Delft](#))
Dissertation: *Next-generation satellite gravimetry for measuring mass transport in the Earth system*
Promotor: [Prof. Dr-Ing. habil. Roland Klees](#)
Supervisor: [Dr. Ir. Pavel Ditmar](#)
- 2004 **M.Sc. in Aerospace Engineering**
Astrodynamics and Space missions ([AS](#)), [TU Delft](#)
Final Thesis: *Numerical Simulation of Launch Vehicles*
Supervisor: [Prof. Ir. B.A.C. Ambrosius](#)
- 2000 **Licenciante in Aerospace Engineering**
Instituto Superior Técnico ([IST](#)), Technical University of Lisbon ([UTL](#))
5th year concluded at [TU Delft](#), through the [ERASMUS program](#)
Report: *Optimum Aerodynamic Shape for a High Altitude Long Endurance Aerostatic Platform*
Supervisor: Prof. Dr. Ir. Theo van Holten

Academic and Work Experience

- Jun. 2018 – present
- Research Engineering/Scientist Associate** at Center for Space Research ([CSR](#)), **UTexas! (UTexas!)**, USA:
- Refinement of the last gravity field solutions from the Gravity Recovery And Climate Experiment (GRACE) mission;
 - Assess signal continuity between GRACE and GRACE-FO;
 - GRACE mission-complete reprocessing;
 - Time-varying gravity fields estimated from Kinematic Orbits;
 - In-house software development in Matlab and Ruby.
 - MSc student supervision and mentoring.

- Aug. 2016 – May 2018 **Postdoctoral Fellow** at Center for Space Research ([CSR](#)), **UTexas!** ([UTexas!](#)), USA:
- Improvements in the calibration of the accelerometers on-board the GRACE satellites, in particular in what relates to temperature effects;
 - Determination of the (non-linear) long-term trends in the GRACE gravity field solutions and their prediction during the GRACE/GRACE-FO gap;
 - Time-varying gravity fields estimated from Kinematic Orbits;
 - In-house software development in Matlab and Ruby.
- Sep. 2011 – Jul. 2016 **Research Associate** at Astrodynamics and Space missions, [TU Delft](#), the Netherlands:
- Calibration of the accelerometers on-board the Swarm satellites;
 - Improvements in the modelling of non-conservative forces acting on satellites;
 - Exploiting Digital Signal Processing (DSP) techniques to merge the measurement of non-gravitational accelerations from different sources: GPS-driven and accelerometer non-gravitational observations;
 - Time-varying gravity fields estimated from Kinematic Orbits;
 - Research project: Assessment of Satellite Constellations for Monitoring the Variations in Earth's Gravity Field;
 - Research project: GOCE+ Theme3: Air density and wind retrieval using Gravity field and steady-state Ocean Circulation Explorer (GOCE) data;
 - Research project: Development of the Swarm Level 2 Algorithms and Associated Level 2 Processing Facility;
 - In-house software development in Fortran, Matlab and Ruby;
 - Student supervision and mentoring.
- Jan. 2007 – Dec. 2015 **PhD Candidate** at [GRS](#), [TU Delft](#), the Netherlands:
- Simulation of future gravimetric satellite missions and noise budget of low-low satellite-to-satellite tracking gravimetric data;
 - Impact of orbit position modelling errors in the quality of satellite gravimetric data;
 - Retrieval of the high-frequency time-variable gravity field of the Earth with numerous satellites;
 - Research project: Assessment of a Next Generation Gravity Mission for Monitoring the Variations of Earth's Gravity Field;
 - Research project: Monitoring and Modelling Individual Sources of Mass Distribution and Transport in the Earth System by Means of Satellites;
 - In-house software development in Fortran and Matlab;
 - Student supervision and mentoring.
- Apr. 2005 – Nov. 2006 **Stress Engineer** at [Global Technics](#), Leiden, the Netherlands:
- Automated design (for weight and stress minimization) of fuselage panels for the Airbus A380 aircraft (in-house implementation of a tool in C++);
 - Trainees supervision and mentoring.
- Oct. 2004 – Jan. 2005 **Aerospace Engineer** at [Delta-Utec](#), Leiden, the Netherlands:
- Contractor Work: Implementation of a Sub-Orbital Optimization Module into the Simulation Tool COLVET (developed in-house at TU Delft).

- Mar. 2004 – Apr. 2004 **Trainee** at the Prins Maurits Laboratorium, [TNO](#), the Netherlands
Supervisor: Ir. Berry Sanders, Rocket Technology Research Group:
- Implementation of the Launch Vehicle Simulation and Optimisation Tool COLVET;
 - Numerical Simulations on Laser Propulsion (appendix of MSc thesis);
 - Collaboration with international colleagues (PT and NL) on a European Space Agency ([ESA](#))-funded project to determine the feasibility of Laser Propulsion.
- Sep. 2001 – Dec. 2001 **Trainee** at European Space Research and Technology Centre ([ESTEC](#)), [ESA](#), Noordwijk, the Netherlands
Supervisor: [Prof. Wubbo Ockels](#):
- Collaboration with fellow MSc colleagues on a space mission design project: *Lunar Exploration with Ariane 5*;
 - Simulation of rocket ascent trajectories (implemented a 2D orbit integrator in Matlab);
 - Optimization of rocket trajectories, thrust and attitude program, fuel consumption and payload;
 - Preliminary lunar mission design.

Awards

- 2017 H2020 Marie Skłodowska-Curie Individual Fellowship [Seal of Excellent](#) to the proposal on [Direct Gravimetric data assimilation into Geophysical models](#)

Invited Presentations

- 2017 Teixeira da Encarnação, J., “Satellite Gravimetry”. In: *Summer School On Data Assimilation and Its Applications In Oceanography, Hydrology, Risk, Safety And Reservoir Engineering*. Sibiu, Romania.
- 2015 Teixeira da Encarnação, J., Arnold, D., Bezdek, A., Dahle, C., Doornbos, E., Ijssel, J. V. D., Jäggi, A., Mayer-gürr, T., Sebera, J., Visser, P., Zehentner, N., “First Monthly Gravity Field Solutions Derived from GPS Orbits of Swarm”. In: *AGU Fall Meeting*. San Francisco, CA, USA: 14-18 Dec.

Collaborations

- 2015 – present International collaboration with Prof. Torsten Mayer-Gürr of the Institute of Geodesy Graz ([IfG](#)) of the Graz University of Technology ([TUG](#)), Dr. Aleš Bezděk of the [ASU](#) of the Czech Academy of Sciences ([AVCR](#)), Prof. Adrian Jäggi of the Astronomical Institute of the University of Bern ([AIUB](#)), Prof. Pieter Visser of the [Aerospace Faculty](#) of the [TU Delft](#) and Prof. C.K. Shum of the School of Earth Science ([SES](#)) of the Ohio State University ([OSU](#)) for the **study of the time-variable gravity field of the Earth estimated from GPS data collected by the Swarm Satellite mission**. These activities have started before we were awarded the funding [ITT posted by the ESA-funded DISC consortium](#) and led to one journal publication in *Advances in Space Research* and yearly contribution to the [EGU](#) General Assembly.
- 2014 – present Collaboration with [TU Delft](#) on the [DopTrack project](#), consisting of a **satellite tracking radio station that exploits the Doppler effect**; co-initiated and promoted the project, secured departmental funding, selected and assembled the hardware, developed software, engaged students and mentored practical undergraduate projects.

Research Projects

- 2022 – 2023 Quantum Space Gravimetry for monitoring Earth’s Mass Transport processes
- 2021 – 2023 NGGM/MAGIC Science Support Study During Phase-A

2021 – 2023	NGGM Phase-A System Study
2020 – 2023	uPGRADE - Miniaturized Prototype for GRavity field Assessment using Distributed Earth-orbiting assets (Funding opportunity: AAC 04/SI/2019 , project nr: 45927)
2019 – present	Multi-approach gravity field models from Swarm GPS data, transfer to operation of production of EGF_SHA_2_ products (DISC contract SW-CO-DTU-GS-111, part of ESA contract 4000109587/13/I-NB)
2017 – 2019	Multi-approach gravity field models from Swarm GPS data (DISC contract SW-CN-DTU-GS-027, part of ESA contract 4000109587/13/I-NB)
2016 – 2019	Gravity Recovery And Climate Experiment (GRACE) Two-year Mission Extension (NASA contract NNL14AA00C)
2013 – 2015	Assessment of Satellite Constellations for Monitoring the Variations in Earth's Gravity Field (ESA contract 4000108663/13/NL/MV)
2013	GOCE+ Theme3: Air density and wind retrieval using GOCE data (ESA contract 400010284/11/NL/EL)
2011 – 2016	Development of the Swarm Level 2 Algorithms and Associated Level 2 Processing Facility (ESA Contract 4000102140/10/NL/JA)
2010	Assessment of a Next Generation Gravity Mission for Monitoring the Variations of Earth's Gravity Field (ESTEC contract 22643/09/NL/AF)
2008	Monitoring and Modelling Individual Sources of Mass Distribution and Transport in the Earth System by Means of Satellites (ESA contract 20403)

Service to Profession

Reviewer for scientific papers submitted to *Advances in Space Research*, *Annales Geophysicae*, *Journal of Geodesy*, *Communications in Nonlinear Science and Numerical Simulation*, *IEEE Geoscience and Remote Sensing Letters*, *International Association of Geodesy Symposia*, *Journal of Geophysical Research: Solid Earth*, *Remote Sensing*, *Sensors and Solid Earth*, cf. [Publons](#)

Extra Training

2017	Dealing with Difficult People, Jeff Stellmach, UTexas!
2017	Conflict Management Foundations, Kimberly Sullivan, UTexas!
2017	Leading without formal authority, Emil Kresl, UTexas!
2017	Meeting effectiveness, Emil Kresl, UTexas!
2015	Scientific Writing, Sören Johnson, TU Delft
1993	Certificate of Proficiency in English
1993	Radio Amateur, call sign CT3IU, class B

Languages

	Speaking	Reading (mother tongue)	Writing
Portuguese			
English	excellent	excellent	excellent
Spanish	good	good	fair
Italian	good	good	fair
Dutch	fair	fair	limited
French	fair	fair	limited

Professional Memberships

since 2012 American Geophysical Union
since 2016 European Geosciences Union